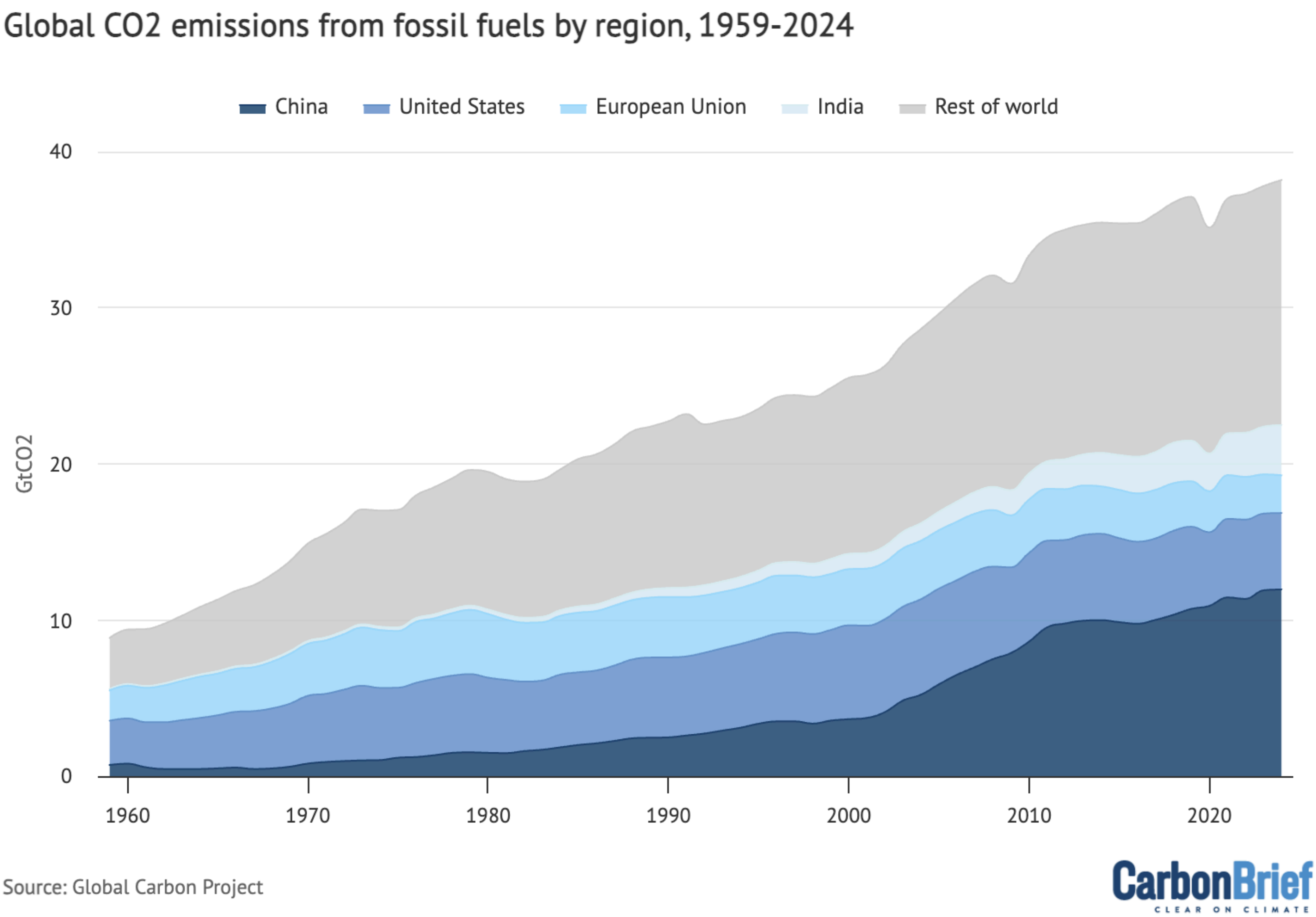
## Visualization




Data source

- News Media URL
- https://globalcarbonbudget.org/gcb-2024/ [National Fossil Carbon Emissions 2024 v1.0]

### 1. What data are visualized?

Annual fossil $CO_2$ emissions (in billion tonnes of $CO_2$ per year, $GtCO_2$) from 1959 to 2024, broken down by five categories: China, India, the United States, the European Union, and Rest of the World. The 2024 value is a preliminary projection, while 1959 - 2023 are confirmed figures. Data are retrieved from the Global Carbon Project's National Fossil Carbon Emissions dataset (GCB2024), with values originally in million tonnes of carbon per year (MtC/yr) converted to $GtCO_2$ by multiplying by 3.664. The chart excludes land-use change emissions and does not adjust for population or trade.

### 2. What does the visualization aim to communicate?

The chart aims to communicate that global fossil $CO_2$ emissions will be reaching a record high of 37.4 $GtCO_2$ in 2024 based on estimates, 0.8% above 2023. The chart is intended to give readers a long-term historical perspective on which countries are responsible for the upward trend. However, the chart largely fails at the regional breakdown goal because of the stacked area format, only China's trend (the bottom layer) is directly readable, while the rest are visually indistinguishable, undermining the chart's own stated purpose.

## 3. What aspects of the visualization do you appreciate?

- The visualization provides a long historical timespan covering 1959–2024 (65 years) gives genuine analytical value where viewers can see structural shifts like China's rapid rise post-2000, the post-Soviet EU decline, and the COVID-19 dip in 2020. Short-term charts might miss all of this.
- The chart has a zero-based y-axis, which is the correct baseline for a stacked area chart showing absolute quantities. This prevents artificial exaggeration of growth, and readers can get an honest sense of the magnitude of total emissions.
- The chart shows a clear overall message. Despite its flaws, the chart successfully conveys the core story: global fossil $CO_2$ emissions have reached a record high, driven primarily by China and the Rest of World.

## 4. What aspects do you find problematic?

- Wrong chart type: A stacked area chart requires readers to judge values by the vertical distance between two moving lines, not from a shared zero baseline. Only China (the bottom layer) is readable accurately, every other country requires mental subtraction across layers, which is error-prone and cognitively demanding. This directly violates the principle that position on a common scale is the most accurate visual encoding
- Total emissions with no per-capita context: China occupies roughly one-third of the chart's visual area, making it appear to bear the most responsibility. However, China's per-capita emissions (~8 t/person) are far below the US (~14 t/person). A general reader with no population awareness would draw the wrong conclusion about individual-level responsibility
- Legend order does not match stacking order. The legend reads: China → United States → European Union → India → Rest of World. But visually in the chart, China is at the bottom and Rest of World dominates the top. A reader naturally tries to match legend order to visual order which might add extra confusion.
- No contextual annotations: The chart spans 65 years but marks no key events like the Paris Agreement (2015), the COVID-19 dip (2020), or China's post-2002 industrialisation surge all appear as unlabelled features that a general reader cannot interpret without prior knowledge
- Use of Color can be improved: China, the United States, European Union, and India are all shown using similar shades of blue, while the Rest of World is grey which reduces separability between the categories. This can be an issue for readers viewing the chart on a small screen or for readers with colour vision difficulties as it is harder to distinguish between the regions.

## 5.How can the visualization be improved?

- Replace the stacked area chart with small multiples (faceted line charts), one panel per region, so each country's trend is readable from a consistent zero baseline without requiring visual subtraction
- Add a parallel per-capita panel alongside the total emissions panel, enabling readers to simultaneously compare aggregate contribution and individual responsibility. Its a fairer basis for policy comparison
- Annotate key events directly on the chart like the Paris Agreement (2015), COVID-19 dip (2020), and the projected year of 1.5°C carbon budget exhaustion so readers can connect data trends to real-world context without external knowledge
- Use color blind friendly palette (e.g. Okabe–Ito ). The palette is designed for scientific figures great for people with various types of color vision deficiency. Recommended in scientific publications

## 6. Who might be a likely target audience for an improved version, and what task or decision would the visualization support for that audience?

The likely target audience for an improved version is the general public, particularly people who casually read climate news and want to understand which countries are most responsible for $CO_2$ emissions. By showing both total emissions and emissions per person side by side, the improved chart gives readers a fairer and more complete basis for comparing countries. This is useful not just for personal understanding but also because how the public perceives national responsibility shapes the pressure governments face to act on climate change.

## 7. Are the data publicly available, or are there suitable surrogate data?

Yes, the data are publicly available and free to download. The main dataset is the National Fossil Carbon Emissions v2024 from the Global Carbon Project, available at globalcarbonbudget.org/carbonbudget as an Excel file. It covers country-level emissions from as far back as 1850 across around 250 countries. To build the per-capita part of the improved chart, population figures can be downloaded separately from the UN World Population Prospects at population.un.org/wpp/downloads, which is also completely free. Together, these two datasets provide everything needed to clean, transform, and rebuild the visualization.